

Preston Rogers

AUTONOMY / ROBOTICS ENGINEER

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Education

Stanford University

M.S. IN ELECTRICAL ENGINEERING

Stanford, CA

Sep 2019 - Jun 2021

- Specialization: Control and Optimization
- Cumulative GPA: 4.1

University of California, Irvine

B.S. IN ELECTRICAL ENGINEERING

Irvine, CA

Sep 2014 - Jun 2019

- Specialization: Digital Signal Processing
- Cumulative GPA: 3.9 (Summa cum laude)

Relevant Academic Projects

GRADUATE SCHOOL

Stanford, CA

Sep 2019 - Jun 2021

- Developed Newton-Raphson inverse kinematics solver for a 5-DOF EMG-controlled robot arm.
- Implemented collision-free path planning for a legged robot in MuJoCo using Sequential Convex Programming with linearized obstacle avoidance constraints.
- Trained a DNN to replicate an MPC trajectory planner for vehicle navigation.

Skills

Programming Languages

- C++, Python, C, MATLAB

Software & Tools

- ROS/ROS2, Docker, Git, Linux, MuJoCo, Gazebo, Altium Designer, PX4, GitLab CI/CD

Technical Domains

- Control Theory, Motion Planning, Autonomous Systems, Convex Optimization

Experience

NASA Jet Propulsion Laboratory

ROBOTICS TECHNOLOGIST

Pasadena, CA

Mar 2023 - Present

- Developed high-fidelity actuator models and authored the controls pipeline section of a peer-reviewed Superlimbs publication.
- Implemented autonomous tube insertion on SRL STS Bradbury Testbed using force/torque-regulated end-effector trajectory control with force-threshold verification.
- Integrated actuator hardware and JSD/Fastcat/FCAT controls stack across multiple robotic platforms within the CASAH software ecosystem.
- Engineered and operated Extended Robust Aerial Autonomy platform, managing Docker containers, telemetry GUI, and ROS1 flight software for GPS/Vicon/VIO flight modes.
- Implemented tension-based PD velocity controller and level-winding system for TYMPO tethered drone spooling mechanism.
- Programmed interruptible trapezoidal motion profiles in embedded C for ERNEST rover actuators and operated rover for terrain traversal experiments in ROS2.
- Performed longitudinal and lateral-directional stability analysis for DARPA EVADE VTOL aircraft in hover and fixed-wing trim configurations.

Dexterity AI

ROBOTICS SOFTWARE ENGINEER

Redwood City, CA

Nov 2021 - Jan 2023

- Developed convex shape estimation pipeline converting 3D RGBD point clouds to 2D package footprints for autonomous grasp planning.
- Implemented multi-strategy pick recovery procedures with gripper reorientation and translational offset fallbacks for failed grasps.
- Increased C++ operational space control test coverage by 20% using GoogleTest, validating motion primitives in simulation.

Glidewell Dental

ELECTRICAL ENGINEERING INTERN

Irvine, CA

Jun 2019 - Sep 2019

- Programmed an embedded microprocessor for use in an X-Ray Scanner.
- Constructed the electrical system incorporating a PLC for a Curing Tunnel.

Prof. Al Faruque Lab (AICPS)

PAID UNDERGRADUATE RESEARCHER

Irvine, CA

Feb 2018 - Mar 2019

- Pioneered a C# GUI for real-time monitoring of sensor values.

Waterborne Skateboards

CO-FOUNDER AND CAD DESIGN ENGINEER

Irvine, CA

Jan 2017 - Jan 2019

- Refined a skateboard swivel truck using Solidworks.
- Forged a partnership with members of the UCI Wayfinder Incubator.

Publications

- E. Ballesteros, P. Rogers, J. Jenkins, K. Carpenter, and H. H. Asada, "Design of Supernumerary Robotic Limbs for the Augmentation of Astronauts Performing Partial-Gravity Extra-Vehicular Activities (EVAs)," *The International Journal of Robotics Research*, 2026.
- M. Dolci, J. Bowkett, P. Bailey, A. Boettcher, J. Kim, P. Rogers, et al., "Robotics Capabilities Development for Mars Sample Return Transfer Activities," *2025 IEEE Aerospace Conference*, pp. 1-25, 2025.
- P. Rogers et al., "A High Fidelity Actuator Model for Robotic Autonomy in Space and Beyond," *in preparation*.

Awards

- NASA JPL Team Certificate Award for outstanding engineering development and demonstration for the ERNEST active suspension vehicle.
- NASA JPL Team Certificate Award for the successful development and demonstration of end-to-end Sample Tube and Lid Transfer using Force Regulation on the SRL STS Bradbury Testbed.